

TCS^AI GOOGLE HACKATHON 2026 · EXECUTIVE BUSINESS CASE



Decision DNA

Business Case & Commercial Opportunity Brief

For: Enterprise Architects, CTOs & TCS Engineering & Digital Practice
Leads

MARKET SEGMENT	ANALYSIS SPEED	INCIDENTS PREVENTED	TCS OPPORTUNITY
Large Enterprise	7.4 sec vs. 3-4 days	70% repeat incidents	₹300Cr+ TAM

THE PROBLEM

Enterprise Amnesia is Costing Millions Per Incident

70%

INCIDENTS ARE
REPEATS OF OLD
MISTAKES

\$847K

COST —
REPRESENTATIVE POST-
MORTEM SYNC OUTAGE
EXAMPLE

3–4 days

MANUAL PRD REVIEW
TIME (VS. 7.4 SEC)

12–18 mo

INSTITUTIONAL
MEMORY "HALF-LIFE"
(WITHOUT DECISION
DNA)

The Root Cause: Enterprise organizations make thousands of consequential decisions — architecture choices, vendor approvals, compliance interpretations, security policies. These accumulate as post-mortems, ADRs, and policy documents that nobody reads. Every new project effectively starts from scratch, unknowingly repeating the same expensive mistakes. This is not a knowledge management problem — it's an enterprise memory problem.

The "Insight to Impact" Latency Problem

When the DataViz PRD team proposed a synchronous REST integration in April 2026, the ADR-018 decision that banned it was made in August 2024. The post-mortem that justified ADR-018 followed a high-impact outage (representative ~\$847K impact). None of the PRD authors knew about ADR-018. It took 3 days of architecture review to catch the violation — 3 days before a sprint started. With Decision DNA, this is caught in 7.4 seconds. Every single time. Automatically.

THE SOLUTION

Enterprise Memory Layer — Temporal Decision Graph

Decision DNA builds a **living, queryable Temporal Decision Graph** from all past decisions — ADRs, post-mortems, security incidents, vendor assessments, and compliance rulings. As new artifacts (PRDs, design docs, code) are submitted, the **Conflict Sentinel multi-agent system** instantly cross-references them against the full institutional memory — surfacing conflicts with grounded evidence before a single line of code is written.

How it works (3 agents): **Decision DNA Orchestrator** routes requests to (1) **Conflict Sentinel** for PRD/design/policy checks and "why" questions, and (2) **Code Review Judge (CRJ)** for code/config/PR reviews. Outputs are structured **Conflict Cards** with evidence citations and a HITL disclaimer.

Powered by: Gemini Enterprise Plus on Vertex AI Agent Engine, grounded via Google Drive Connector + Direct Uploads against a curated institutional memory corpus (ADR, post-mortems, security policies, vendor guidelines).

Core Value Proposition: Every governance document becomes a "learning" that permanently protects the organization. A security incident documented in 2022 will prevent the same mistake in a 2026 PRD. The organizational intelligence grows cumulatively — unlike human memory, it never fades, transfers when people leave, and scales infinitely.

Scenario	Without Decision DNA	With Decision DNA	Value
PRD Review	3–4 days manual architecture board	7.4 sec automated conflict scan	99.8% faster
GDPR Gap Detection	External audit, ₹50L+ fee	Instant policy conflict flag	Pre-emptive
Repeat Incident Prevention	70% of incidents are documented repeats	70% caught at PRD stage	₹Cr+ saved/incident
Vendor Risk	Legal sends email 2 weeks later	Immediate unapproved vendor alert	Pre-build
Knowledge Retention	Decisions walk out the door when people leave	100% institutional memory retained	Zero knowledge drain

COMMERCIAL VALUE SUMMARY

₹5–15Cr

Avg. incident cost prevented / year

5,000x

Faster conflict detection

100%

Grounded citations — always auditable

TCS Commercial Opportunity

- 1

Architecture Governance Platform: License Decision DNA as an enterprise architecture governance tool. Target TCS's 200+ large enterprise clients (Fortune 1000 / BFSI / Government). ₹80L–1.5Cr ARR per client. 30-client target = ₹24–45Cr ARR by Year 2.
- 2

Digital Transformation Risk Gating: Embed Decision DNA into TCS's large-scale digital transformation programs as a mandatory governance checkpoint — preventing costly mid-project compliance failures and scope changes. Reduces project overruns and protects TCS's own margins.
- 3

TCS Delivery Quality Shield: Deploy internally across TCS's delivery centers and practice units. Every client deliverable (designs, PRDs, architecture docs) automatically screened against TCS's accumulated institutional knowledge base of 10,000+ past decisions.
- 4

Compliance AI Premium: For regulated industries (Banking, Pharma, Government), position Decision DNA as a mandatory compliance assurance layer — ensuring every design decision is pre-screened against applicable regulatory frameworks (GDPR, DPDPA, RBI IT Framework).

Differentiation vs. Confluence / Jira Knowledge Bases: These tools require humans to search for, find, and apply past decisions. Decision DNA is *proactive* — it automatically brings relevant historical context to every new artifact without requiring engineers to know what to search for. The system catches what humans forget to check.